Data Management and Sharing Plan

1 Roles and Responsibilities The Principal Investigator (PI), Dr. Rebecca Napolitano, will have primary responsibility for data management throughout the project lifecycle and will ensure data is collected, stored, preserved, and shared in accordance with this plan and NSF policies. Co-PI Wauthier is responsible for SAR data products, including ICEYE-derived building-level inundation measurements and Sentinel-1 flood-extent imagery. Co-PI Busse is responsible for hydrological model inputs, outputs, and associated metadata. All three PIs are at Penn State.

2 Expected Data and Formats This project will generate the following data types:

ICEYE commercial SAR data (restricted): Building-level flood depth and inundation duration measurements for 271 downtown Montpelier buildings across the 2023 and 2024 flood events, acquired under an NDA data access agreement with ICEYE. Raw ICEYE imagery and building-level measurements cannot be redistributed under the data access agreement; derived products (per-building duration statistics, validation metrics, anonymized scatter plots) will be shared. Formats: CSV summary tables (.csv), PDF figures (.pdf).

Sentinel-1 SAR flood-extent products: Amplitude imagery and flood-extent classifications for Montpelier, VT across both flood events, used for broader spatial context and cross-checking of ICEYE building-level classifications. Formats: GeoTIFF (.tif), CSV summary tables (.csv).

NOAA precipitation and forecast inputs: NOAA precipitation data and forecast products used for pre-storm staging and mass balance model inputs. Formats: CSV (.csv), NetCDF (.nc).

Hydrological mass balance model outputs: Drainage timescale estimates, sub-basin parameterization, watershed routing outputs, and uncertainty envelopes for the Winooski watershed. Formats: NetCDF (.nc), CSV (.csv).

Bayesian posterior distributions: Posterior duration distributions per building, credible intervals, and error model parameters. Formats: NetCDF (.nc), CSV summary tables (.csv).

Fragility surface parameters: Duration-depth fragility surfaces for pre-code load-bearing masonry archetypes, calibrated on 2023 observations and validated against 2024. Formats: CSV (.csv), Python serialization (.pkl).

Field damage observations: Building-level damage state classifications (no damage, cosmetic, structural) from field assessment of 271 downtown Montpelier buildings. These data are building-identifiable; anonymized versions with parcel-level identifiers removed will be shared publicly, and the full geo-referenced dataset will be available to qualified researchers under a data use agreement. Formats: CSV (.csv), GeoJSON (.geojson).

Community co-development workshop materials: Workshop documentation, structured elicitation results, participant feedback, and facilitation guides. Formats: PDF (.pdf), anonymized transcripts (.txt, .docx).

Decision framework outputs: Retrofit priority rankings, what-if simulation outputs, and community-scale drainage versus individual building retrofit comparisons. Formats: CSV (.csv), PDF reports for community partners.

Dissemination materials: Publications, presentations, webinars (.pdf, .pptx, .mp4).

Metadata will include instrument specifications (SAR sensor, orbit, acquisition dates), data processing parameters, model version information, and field data collection protocols.

3 Access, Sharing, and Reuse Policies SAR-derived flood-extent products, mass balance outputs, posterior distributions, and fragility surface parameters will be shared via DesignSafe-CI under open-access licenses upon publication or within 12 months of data collection completion. ICEYE commercial SAR data is acquired under a data access agreement that restricts redistribution of raw imagery; derived per-building statistics and validation metrics will be shared on DesignSafe-CI. Bayesian fusion, mass balance parameterization, and decision framework code will be released on GitHub under an MIT open-source license. Field damage observations will be shared on the Open Science Framework: anonymized data publicly, geo-referenced data under a data use agreement. Community workshop materials will be anonymized before sharing; access to identifiable materials will be restricted to project personnel. Publications and presentations will be made openly available via institutional repositories.

The hazard-related data and metadata will be archived on the Data Depot at DesignSafe-CI (the standard repository for the natural hazards community), following established standards listed on the DesignSafe-CI website and subject to DesignSafe-CI Data Sharing and Archiving Policies.

4 Reproducibility All data processing and analysis scripts will be version-controlled using Git and stored on GitHub with appropriate documentation. Computational environments will be documented using Docker containers or Conda environment files, with configuration files shared alongside data. All statistical analyses and visualizations will include code to reproduce them from processed data. Open-source software tools will be prioritized to remove licensing barriers.

5 Data Storage, Preservation and Backup All data and code will be archived on DesignSafe-CI. PI Napolitano is an affiliate of Penn State's Institute for Computation and Data Science (ICDS) and, through its Advanced Cyber Infrastructure (ICDS-ACI), has access to active and near-line storage that is backed up daily. Co-PIs Wauthier and Busse will use ICDS-ACI storage for active research data, ensuring all project data resides within Penn State's backed-up infrastructure.

6 Ethics and Privacy The project will comply with all relevant regulations, including Institutional Review Board (IRB) requirements for human subjects research. Community co-development workshop participants will be fully informed of data collection and sharing plans. Workshop notes will be anonymized before sharing; personally identifiable information will be protected using encryption and access controls. Field damage observations are building-identifiable; publicly shared versions will have parcel-level identifiers removed, and geo-referenced data will be available only under a data use agreement to protect building owner privacy. This plan will be reviewed annually and revised as needed in consultation with the Penn State IRB.